

**BEFORE YOU READ — ANSWER FROM MEMORY**

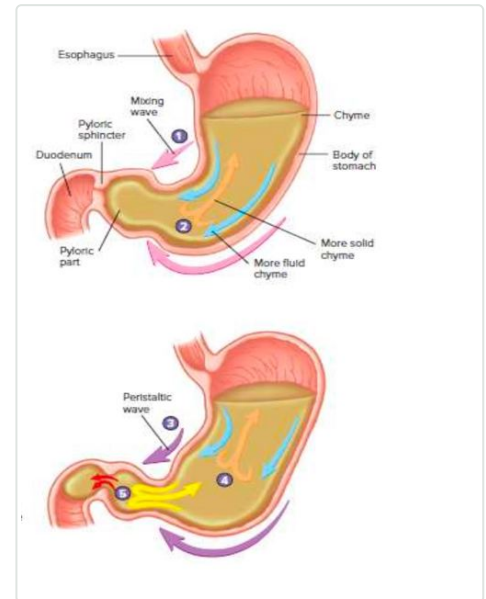
1. Which part of the stomach stores food, and which part mixes it and empties it into the duodenum?
2. Which cell secretes HCl, which secretes pepsinogen, and which makes gastrin?
3. What are the three phases of acid secretion, and which contributes the most?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Stomach: functions and movements

Region	Wall and contractions	Job
Proximal (fundus + body)	Thin wall; contracts weakly and infrequently	Stores large volumes via receptive relaxation
Distal (pylorus)	Thick wall; strong, frequent peristaltic contractions	Mixes and propels food into the duodenum; gastric emptying

- **Mixing wave:** fluid chyme moves toward the pyloric sphincter while the more solid centre is squeezed back toward the body.
- **Peristaltic waves** go the same way but are **stronger**: only a few mL of mostly fluid chyme pass into the duodenum; the rest is forced back for **further mixing**.



Mixing wave (top) and peristaltic wave (bottom).

Main function	Detail
Store + mechanical breakdown	Bulk storage of undigested food; mechanical breakdown
Chemical breakdown	Disrupts chemical bonds with acid and enzymes (pepsin)
Intrinsic factor	Aids vitamin B12 absorption
Absorption / endocrine	Very little nutrient absorption (some drugs are absorbed); enteroendocrine cells make hormones (gastrin)

2 Gastric glands: who secretes what

Cell	Secretes	Function
Mucous neck cell	Mucus; bicarbonate	Physical barrier between lumen and epithelium; buffers acid to protect the epithelium
Parietal cell	HCl; intrinsic factor	HCl activates pepsin and kills bacteria; intrinsic factor complexes with vitamin B12 to permit absorption
ECL cell	Histamine	Stimulates gastric acid secretion
Chief cell	Pepsinogen ; gastric lipase	Pepsin digests proteins; gastric lipase digests fats
D cell	Somatostatin	Inhibits gastric acid secretion
G cell	Gastrin	Stimulates gastric acid secretion

Mucus	Role
Soluble	Mixes with stomach contents and lubricates chyme
Insoluble	Forms a protective barrier against the high acidity of the stomach contents

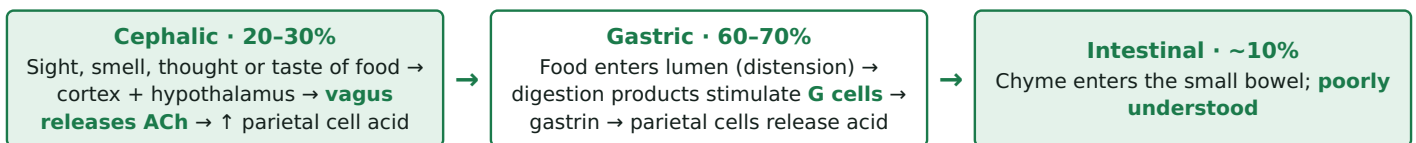
GI secretions total ~6–8 L/day and serve to lubricate (water, mucus), protect (mucus), sterilise (HCl), neutralise (HCO_3^-) and digest (enzymes).

3 Functions of gastric HCl

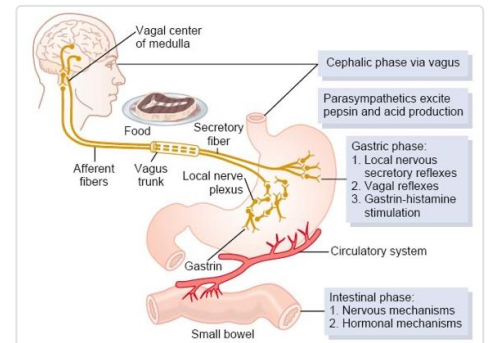
1	Activates pepsinogen into pepsin	4	Kills bacteria in food
2	Gives the optimum pH for pepsin	5	Helps Fe^{2+} and Ca^{2+} absorption
3	Denatures protein → helps its digestion	6	Promotes pancreatic, small-intestinal and bile secretion

4 Three phases of acid secretion

VISUAL ANCHOR · CEPHALIC → GASTRIC → INTESTINAL



- Gastric phase begins with **gastric distension**; **distension alone** can increase acid production.
- It lasts **until the stomach is empty**.
- Regulation: cephalic phase via the **vagus**; gastric phase via **local nervous reflexes, vagal reflexes and gastrin-histamine** stimulation; intestinal phase via **nervous and hormonal** mechanisms.



Regulation of gastric secretion.

5 Gastric hormones

Hormone	Made by	Released by	Action
Gastrin	G cells of the antrum	Amino acids, protein, vagal tone, antral distension, GRP, pH > 3.0, alcohol, histamine. Inhibited by pH < 3.0, somatostatin, secretin, CCK, VIP, GIP, glucagon	Targets parietal + chief cells : ↑ acid (direct, and potentiates histamine); ↑ water/electrolytes from stomach, pancreas, liver, Brunner's glands; ↑ motility; ↑ mucosal growth
Somatostatin	CNS, antrum, fundus, SI, colon; pancreatic D cells	Antral acidification; fats, protein and acid in the duodenum	"Master off switch" : inhibits most GI hormones, GI + pancreatic secretion and intestinal motility. Clinical: octreotide ↓ fistula output, oesophageal variceal bleed, endocrine tumour symptoms
GRP	Gastric antrum, small-bowel mucosa	Vagal stimulation	Stimulates release of all GIT secretions and motility
Histamine	ECL cells, mast cells	Gastrin, ACh, epinephrine. Inhibited by somatostatin	Stimulates parietal cells

The stimulus for hormone release is **food**; its composition sets the timing and which hormone is released. Messengers act as endocrine, paracrine, autocrine or neurocrine signals.

6 Duodenum and small-intestinal secretion

Feature	Detail
Receives	Chyme from the stomach; bile (liver, gallbladder) and pancreatic juice
Regulates	The rate of gastric emptying and triggers hunger signals , via hormones from the duodenal epithelium
Hormones	Secretin and cholecystokinin — both encourage secretion of bile and pancreatic juice
Absorbs	Nutrients, even more than the stomach → bypassed in gastric bypass surgery for obesity to ↓ absorption
Secretion	Duodenal + intestinal glands, 1-3 L/day : ions, mucoprotein, IgA, enzymes (e.g. enterokinase). Roles: protection by mucus; digestion (peptidase, sucrase, lipase); dilution

P Must-remember pearls

- **Proximal stomach stores** (thin wall, receptive relaxation); **distal stomach mixes and empties** (thick wall, strong peristalsis).
- **Parietal** = HCl + intrinsic factor; **chief** = pepsinogen + gastric lipase; **ECL** = histamine; **G** = gastrin (stimulates acid); **D** = somatostatin (inhibits acid).
- Acid secretion: **cephalic 20-30% (vagus, ACh)**, **gastric 60-70% (distension, gastrin)**, **intestinal ~10%**. Distension alone can raise acid.
- Gastrin is inhibited by **pH < 3.0**, somatostatin, secretin and CCK; **somatostatin is the “master off switch”**; histamine and gastrin both stimulate parietal cells.
- Insoluble mucus is the **acid barrier**; HCl **activates pepsinogen** and helps Fe²⁺/Ca²⁺ absorption. The **duodenum** absorbs more than the stomach and sets the gastric emptying rate.

Q Test yourself

1. After a large meal, which part of the stomach holds the food, and how does it manage this?

- A. The fundus and body, which have a thin wall, contract weakly and relax to accept large volumes
- B. The pylorus, which contracts strongly and frequently to hold the meal
- C. The duodenum, which relaxes to receive the whole meal at once
- D. The fundus and body, which contract strongly and frequently to mix the meal
- E. The pylorus, which relaxes to store food and then contracts weakly

2. A patient has a profound loss of gastric acid. Which function is directly lost?

- A. Receptive relaxation of the fundus
- B. Secretion of mucus by mucous neck cells
- C. Secretion of gastrin by G cells
- D. Activation of pepsinogen into pepsin
- E. Release of somatostatin by D cells

3. A cook smells food before eating and her gastric acid output rises. Which pathway explains this?

- A. Distension of the gastric lumen stimulates G cells to release gastrin
- B. Somatostatin from D cells stimulates the parietal cells
- C. Cortex and hypothalamus signal the vagus to release ACh, which raises parietal cell acid production
- D. Histamine from mast cells is released by falling gastric pH
- E. Chyme entering the small bowel triggers hormonal release

4. Which secretion forms a protective barrier against the high acidity of the stomach contents?

- A. Soluble mucus
- B. Insoluble mucus
- C. Pepsinogen
- D. Gastric lipase
- E. Intrinsic factor

5. A patient loses the cells that secrete both HCl and intrinsic factor. Which nutrient absorption is impaired as a result of the intrinsic factor loss?

- A. Sodium
- B. Free amino acids
- C. Long-chain fatty acids
- D. Glucose
- E. Vitamin B12

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A — Proximal stomach (fundus + body): thin wall, weak infrequent contractions, receptive relaxation → stores large volumes.
- 2 — D — HCl activates pepsinogen into pepsin and provides the optimum pH for pepsin.
- 3 — C — Cephalic phase (20–30% of acid): sight/smell/thought/taste → cortex + hypothalamus → vagal ACh → parietal cells.
- 4 — B — Insoluble mucus forms the protective barrier; soluble mucus mixes with contents and lubricates chyme.
- 5 — E — Parietal cells secrete intrinsic factor, which complexes with vitamin B12 to permit absorption.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam